

FAIRFIELD COUNTY AP, OH, USA

WMO: 724294

Lat: 39.756N

Lon: 82.657W

Elev: 265

StdP: 98.18

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 53844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.0	-20.9	0.6	-15.7	-17.5	0.8	-12.4	10.6	2.7	9.2	2.8	1.3	260	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.4	23.4	31.2	22.9	29.9	22.2	25.1	30.4	24.2	28.9	23.5	27.9	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.7	27.8	22.6	17.9	26.9	22.1	17.4	26.3	77.3	30.2	74.1	29.1	71.0	27.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
8.9	7.9	7.1	DB	-21.8	34.0	4.6	1.6	-25.1	35.2	-27.8	36.1	-30.4	37.0	-33.8	38.2	
			WB	-22.3	26.3	4.6	0.9	-25.6	26.9	-28.3	27.4	-30.8	27.9	-34.2	28.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-1.5	0.2	5.1	11.5	17.1	21.4	22.9	22.0	18.5	12.1	6.0	1.2
	DBStd	10.01	7.21	6.56	6.08	5.03	4.49	3.32	2.86	2.82	3.98	5.01	5.25	5.89
	HDD10.0	1319	362	281	177	41	3	0	0	0	1	34	142	279
	HDD18.3	2988	616	509	411	209	78	11	2	4	45	202	370	531
	CDD10.0	1847	5	6	26	87	224	343	400	373	257	99	22	6
	CDD18.3	473	0	0	1	5	41	104	144	118	51	9	0	0
	CDH23.3	4523	0	1	12	68	413	958	1308	1102	561	99	2	0
(14) Wind	WSAvg	2.8	3.5	3.4	3.5	3.5	2.9	2.5	2.0	1.8	1.9	2.6	3.0	3.2
	PrecAvg	1036	76	57	87	98	107	105	107	89	87	74	69	80
	PrecMax	1360	203	110	192	200	212	213	240	193	163	156	132	190
	PrecMin	767	30	26	40	26	25	23	58	32	31	24	21	34
	PrecStd	137	37	23	32	35	50	43	40	36	35	31	29	31
	DB	18.0	20.1	25.2	27.6	31.3	32.9	33.9	33.6	32.9	29.1	22.6	18.6	
	MCWB	13.9	14.7	16.6	18.3	21.5	22.6	24.7	23.5	22.2	20.8	15.8	15.2	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	15.1	16.3	21.6	25.5	29.1	31.4	32.2	32.0	30.9	26.1	19.7	15.5	
	2% MCWB	12.5	11.9	14.5	16.4	20.3	22.6	24.3	23.3	21.9	18.9	14.1	12.2	
	5% DB	12.4	13.1	18.3	23.2	27.7	30.0	31.0	30.2	28.6	23.6	17.5	12.9	
	5% MCWB	10.3	9.8	12.7	15.3	19.8	21.9	23.5	22.7	20.7	17.4	13.0	10.4	
	10% DB	8.9	10.0	15.8	21.1	25.9	28.5	29.2	28.7	26.9	21.2	15.1	10.3	
	10% MCWB	6.3	6.8	11.4	14.4	19.2	21.3	22.7	22.0	19.7	16.3	11.4	8.0	
	WB	15.1	15.5	17.5	20.2	23.3	25.0	26.6	25.7	24.4	22.1	17.7	15.6	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	17.3	19.3	24.2	25.9	29.0	30.5	31.4	31.0	29.9	27.6	20.4	18.1	
	2% WB	13.0	13.0	15.4	18.0	21.9	23.9	25.4	24.8	23.1	20.4	15.5	13.0	
	2% MCDB	15.1	15.4	19.9	22.9	26.9	29.1	30.8	29.8	27.6	24.1	18.5	14.7	
	5% WB	10.4	10.1	13.8	16.6	20.8	23.0	24.4	23.9	22.1	18.6	13.6	10.7	
	5% MCDB	12.0	12.9	17.4	21.0	25.8	28.2	29.2	28.2	26.5	22.2	16.7	12.6	
	10% WB	6.7	6.9	11.8	15.3	19.8	22.2	23.6	23.0	21.0	16.9	11.8	7.9	
	10% MCDB	8.6	9.8	15.4	19.8	24.6	27.0	27.9	26.9	24.9	20.5	14.8	9.9	
(35) Mean Daily Temperature Range	MDBR	9.0	9.8	10.9	12.9	12.1	11.5	11.4	12.3	13.5	12.7	11.0	8.5	
	5% DB	10.9	13.8	15.4	15.6	14.5	13.5	12.7	13.9	15.5	15.4	14.5	11.6	
	MCWBR	9.2	10.6	9.6	8.7	7.1	6.0	5.8	6.5	7.3	8.8	10.0	9.6	
	5% WB	10.1	12.8	12.9	12.8	12.0	11.0	10.9	11.1	12.2	12.8	11.8	10.4	
	MCWBR	9.1	10.6	9.0	8.1	6.5	5.5	5.5	5.7	6.3	7.8	9.2	9.5	
(40) Clear-Sky Solar Irradiance	taub	0.309	0.326	0.355	0.392	0.436	0.456	0.466	0.458	0.399	0.365	0.326	0.304	
	taud	2.442	2.368	2.361	2.276	2.183	2.171	2.154	2.156	2.301	2.393	2.472	2.479	
	Ebn at Noon	858	890	899	882	847	828	815	810	840	834	831	838	
	Edh at Noon	81	99	112	130	146	148	149	145	118	96	78	72	
(44) All-Sky Solar Radiation	RadAvg	1.65	2.46	3.50	4.73	5.47	5.99	5.79	5.41	4.43	3.08	2.10	1.41	
	RadStd	0.16	0.33	0.32	0.42	0.46	0.41	0.32	0.30	0.41	0.28	0.28	0.16	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+158	N/A
		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	+130	+71

Nomenclature: See separate page